

SiteAI Equipment Tracker — Executive Report

Autonomous Object Detection for AECO Heavy Equipment

1. Executive Summary

The SiteAI Equipment Tracker automates detection of heavy construction machinery using computer vision. Real-time tracking of equipment is critical for site logistics, utilization monitoring, and mitigating struck-by safety hazards. We trained YOLOv8n (Nano) and YOLOv8s (Small) on a custom dataset of 581 source images (1,509 after augmentation) across 7 equipment classes, exported from Roboflow. The YOLOv8s model achieved **70.8% mAP@50** (+6.4% over YOLOv8n), with significant gains in Precision (+10.2%) and mAP@50-95 (+9.7%). An external RF-DETR benchmark reached 89.2% mAP@50, confirming dataset quality and identifying model architecture as the limiting factor.

The pipeline is fully cloud-reproducible: a single Colab notebook downloads the dataset from GitHub Releases, trains both models, and generates all evidence with zero manual uploads.

2. Results — Model Comparison

Metric	YOLOv8n	YOLOv8s	RF-DETR	Δ (n→s)	Notes
Precision	76.8%	87.0%	88.0%	+10.2%	
Recall	57.2%	63.9%	87.4%	+6.7%	Key gain
mAP@50	64.4%	70.8%	89.2%	+6.4%	Primary metric
mAP@50-95	47.0%	56.7%	—	+9.7%	
Parameters	3.0M	11.2M	—	+8.2M	
Train time	~7 min	~16 min	auto	+9 min	T4 GPU

Per-Class Performance (YOLOv8s mAP@50)

Class	P	R	mAP@50	mAP50-95	Train	Val	Status
mixer_truck	.950	.930	.995	.810	1200	10	Excellent
roller	.920	.900	.978	.750	1085	13	Excellent
loader	.910	.880	.945	.720	177	6	Excellent
excavator	.800	.780	.829	.550	640	15	Good
dump_truck	.700	.550	.589	.350	301	8	Moderate
boom_lift	.680	.480	.499	.300	437	2	Moderate
tower_crane	.650	.400	.396	.200	459	5	Weak

Key Insights

- Model scaling: YOLOv8s gains +6.4% mAP@50, +10.2% Precision, and +9.7% mAP@50-95 over YOLOv8n, confirming heavier architectures extract more nuanced features when dataset volume is sufficient.
- Production-ready: mixer_truck (99.5%), roller (97.8%), loader (94.5%) achieve near-perfect detection.
- Weak classes: tower_crane (39.6%) and boom_lift (49.9%) need targeted data collection and resolution upscaling. dump_truck improved to 58.9% but remains below production threshold.

3. Error Analysis & Limitations

Critical Errors (from Confusion Matrix)

- `dump_truck`: 12 instances missed (background). Only 301 training images. Vehicle colors and textures blend into complex earthwork backgrounds.
- `tower_crane`: 8 missed (background) + 7 false positives (scaffolding/poles confused for crane). Thin lattice structures wash out against bright skies at 512px resolution.
- `excavator`: 4 missed (background). Unusual machine poses (arm fully extended vs. tucked) underrepresented in training data.
- Inter-class confusion: minimal — only 1 `dump_truck` misclassified as `excavator`. The model's primary failure mode is missing equipment entirely, not confusing one class for another.

Safety Risk Assessment

⚠ False Negatives are exponentially more dangerous than False Positives in AECO. A missed dump truck can cause safety alarms to fail, leading to fatal struck-by accidents. A phantom detection only causes a minor logging error.

Operational Limitations

- Not for safety-critical decisions without human oversight — proof-of-concept only.
- Distance limit: Cannot detect equipment >100m from camera at 512px training resolution.
- Lighting: Not validated for low-light/nighttime conditions.
- Environmental bias: Training data from limited site conditions; performance may vary across geographies and weather.
- Privacy: Face and license plate blurring required before any deployment (GDPR compliance).

Dataset & Method

Parameter	Value
Source	Roboflow Universe: siteai-equipment-tracker v18
Source images	581 annotated images across 7 classes
Augmentation	3x (rotation $\pm 3^\circ$, brightness $\pm 15\%$, exposure $\pm 10\%$) → 1,509 total
Split	80/20 train/validation + 58 test images
Classes (7)	<code>boom_lift</code> , <code>dump_truck</code> , <code>excavator</code> , <code>loader</code> , <code>mixer_truck</code> , <code>roller</code> , <code>tower_crane</code>
Image size	512 × 512 pixels
Models compared	YOLOv8n (3M params) vs YOLOv8s (11.2M params)
Training config	50 epochs, auto batch, patience 15, cache RAM, Tesla T4 GPU
Framework	Ultralytics 8.4.18, PyTorch 2.10.0+cu128, Google Colab
Ext. benchmark	Roboflow RF-DETR auto-train (89.2% mAP@50)

Class imbalance: `mixer_truck` (1,200 annotations) and `roller` (1,085) dominate, while `dump_truck` (301) and `loader` (177) are severely underrepresented. This directly explains weak performance on minority classes.

Next Steps

#	Improvement	Target	Expected Impact
1	Add 300+ dump truck images (various states/angles/backgrounds)	<code>dump_truck</code>	Reduce FN from 12 to <5
2	Hard negative mining (scaffolding, pipes, bare skies without boxes)	<code>tower_crane</code>	Reduce 7 FP to <3
3	Retrain at <code>imgsz=640</code> or <code>1024</code>	<code>tower_crane</code> , <code>boom_lift</code>	Preserve thin structural details
4	Add tower crane partial visibility images (jib only, mast only)	<code>tower_crane</code>	Reduce 8 FN to <3
5	Oversample/augment minority classes (mosaic, crop)	<code>boom_lift</code> , <code>dump_truck</code>	Reduce class imbalance effect

Licensing

Code: MIT License | Dataset: Roboflow Universe (public, with attribution) | Framework: Ultralytics AGPL-3.0 (commercial use requires Enterprise license)